

Government Bus Crowd Prediction System

Combined Presentation Content + Prototype Website Code

PART 1 — PRESENTATION CONTENT

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GOVERNMENT BUS CROWD PREDICTION SYSTEM

Machine Learning Based Web Solution

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GOVERNMENT BUS CROWD PREDICTION SYSTEM

Machine Learning Based Web Solution

PROBLEM

- Peak-hour overcrowding
- Difficult manual estimation
- Passengers need advance crowd information

SOLUTION

- Web-based prediction system
- Uses bus, time, day and passenger data
- Gives instant crowd status

BENEFIT

- Better passenger planning
- Simple decision support
- Can scale to more buses/routes

INPUT DATA

PROCESS / ML

CROWD SCORE

RESULT

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Slide 3

TECHNICAL APPROACH

METHODOLOGY / WORKFLOW

1 BUS INPUT

2 TIME + DAY

3 PASSENGER DATA

4 SCORE PROCESSING

5 CROWD DECISION

DECISION LOGIC

Peak hours increase score

Monday / Friday increase score

HISTORY FACTOR

Previous passenger count above threshold increases score

FINAL OUTPUT

■ HIGH ≥ 90 ■ MEDIUM 55–89 ■ LOW < 55

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WEBSITE WORKING

USER INPUT → PREDICTION → RESULT

1. USER INPUT

Bus Number → Select 1–6

Bus Time → Select departure time

Day → Monday–Sunday

Previous Passengers → Enter count

Expected Passengers → Enter count

2. PREDICT CROWD

Click “Predict Crowd”

System processes the entered values.

Peak-hour time, selected day and previous passenger count affect the crowd score.

The current prototype uses demo prediction logic and is ready for future ML API integration.

3. RESULT

■ HIGH CROWD

High expected crowd

■ MEDIUM CROWD

Medium expected crowd

■ LOW CROWD

Low expected crowd

Result also shows bus, time, day and passenger values.

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Slide 5

FEASIBILITY, IMPACT & FUTURE SCOPE

TECHNICAL FEASIBILITY

- HTML, CSS and JavaScript
- Easy ML/API integration
- Responsive browser interface

OPERATIONAL VALUE

- Quick crowd prediction
- Helps passengers plan
- Supports transport monitoring

ECONOMIC VALUE

- Low-cost digital approach
- Reduces manual estimation effort
- Reusable for multiple routes

IMPACT

- Better awareness of expected bus crowd
- Supports smarter public transport decisions
- Useful for passengers, staff and transport authorities

FUTURE ENHANCEMENT

- Train with real passenger data
- Add route/stop information
- Live passenger-count / GPS integration
- Dashboard + high-crowd alerts
- Continuous model improvement

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PART 2 — PROTOTYPE WEBSITE CODE

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Government Bus Crowd Prediction</title>
<style>
*{box-sizing:border-box}
body{
margin:0;
font-family:Arial,Helvetica,sans-serif;
background:linear-gradient(135deg,#071a2b 0%,#0b2f3f 50%,#08261f 100%);
color:#f5fbff;
}
.container{
max-width:900px;
margin:auto;
padding:45px 24px 70px;
}
h1{
font-size:34px;
margin:0 0 8px;
font-weight:750;
color:#e9fbff;
text-shadow:0 2px 12px rgba(0,220,200,.18);
}
.subtitle{
color:#b9d7df;
font-size:15px;
margin-bottom:32px;
}
hr{
border:0;
border-top:1px solid #24515c;
margin:28px 0;
}
label{
display:block;
font-size:14px;
color:#d8edf2;
margin:16px 0 8px;
}
select,input{
width:100%;
padding:13px 14px;
border:1px solid #28606b;
border-radius:10px;
background:#102f3a;
color:#f5fbff;
font-size:15px;
outline:none;
box-shadow:0 4px 16px rgba(0,0,0,.12);
}
select:focus,input:focus{
border-color:#25c7b5;
box-shadow:0 0 3px rgba(37,199,181,.14);
}
.row{
display:grid;
grid-template-columns:1fr 1fr;
gap:18px;
}
button{
width:100%;
margin-top:28px;
padding:14px;
border-radius:10px;
border:1px solid #20b8aa;
background:linear-gradient(135deg,#0f8f86,#176c91);
color:#fff;
font-size:15px;
font-weight:600;
cursor:pointer;
box-shadow:0 8px 22px rgba(10,120,120,.25);
transition:transform .15s ease,filter .15s ease;
}
button:hover{
background:linear-gradient(135deg,#13a49a,#1b7fa7);
filter:brightness(1.08);
transform:translateY(-1px);
}
.result{
margin-top:18px;
padding:16px 18px;
border-radius:10px;
display:none;
box-shadow:0 8px 24px rgba(0,0,0,.16);
}
.result.low{
display:block;
```

Prototype Website Code (continued)

```
        background:#0d4b3b;
        border:1px solid #1aa879;
    }
    .result.medium{
        display:block;
        background:#55430c;
        border:1px solid #b58b19;
    }
    .result.high{
        display:block;
        background:#581f2b;
        border:1px solid #b63852;
    }
    .badge{
        font-weight:700;
        font-size:16px;
    }
    .desc{
        margin-top:12px;
        color:#eef8fa;
    }
    .info{
        margin-top:14px;
        background:#123d55;
        padding:13px 15px;
        border-radius:8px;
        color:#c8e5f2;
        font-size:14px;
    }
    .footer{
        color:#87aab4;
        font-size:13px;
        margin-top:18px;
    }
    @media(max-width:650px){
        .row{grid-template-columns:1fr}
        h1{font-size:28px}
        .container{padding-top:28px}
    }
</style>
</head>
<body>
<div class="container">
    <h1>■ Government Bus Crowd Prediction</h1>
    <div class="subtitle">Predict the expected crowd level of a government bus using Machine Learning.</div>

    <hr>

    <div class="row">
        <div>
            <label for="bus">Bus Number</label>
            <select id="bus">
                <option>1</option><option>2</option><option>3</option>
                <option>4</option><option>5</option><option>6</option>
            </select>
        </div>
        <div>
            <label for="time">Bus Time</label>
            <input id="time" type="time" value="08:15">
        </div>
    </div>

    <label for="day">Day</label>
    <select id="day">
        <option>Monday</option><option>Tuesday</option><option>Wednesday</option>
        <option>Thursday</option><option>Friday</option><option>Saturday</option><option>Sunday</option>
    </select>

    <label for="previous">Previous Passengers</label>
    <input id="previous" type="number" min="0" value="70">

    <label for="expected">Expected Passengers</label>
    <input id="expected" type="number" min="0" value="78">

    <button onclick="predictCrowd()">■ Predict Crowd</button>

    <div id="result" class="result">
        <div id="badge" class="badge"></div>
        <div id="desc" class="desc"></div>
        <div id="info" class="info"></div>
    </div>

    <div class="footer">Government Bus Crowd Prediction System • Machine Learning Prototype</div>
</div>

<script>
function predictCrowd(){
    const bus = document.getElementById("bus").value;
    const time = document.getElementById("time").value;
    const day = document.getElementById("day").value;
```

Prototype Website Code (continued)

```
const previous = Number(document.getElementById("previous").value);
const expected = Number(document.getElementById("expected").value);

let hour = Number(time.split(":")[0]);
let score = expected;

// Demo prediction logic. Replace this section with your ML API later.
if((hour >= 7 && hour <= 9) || (hour >= 16 && hour <= 19)) score += 15;
if(day === "Monday" || day === "Friday") score += 8;
if(previous > 60) score += 5;

let level, cls, message;
if(score >= 90){
  level = "■ HIGH CROWD";
  cls = "high";
  message = "This bus is expected to have a high crowd.";
}else if(score >= 55){
  level = "■ MEDIUM CROWD";
  cls = "medium";
  message = "This bus is expected to have a medium crowd.";
}else{
  level = "■ LOW CROWD";
  cls = "low";
  message = "This bus is expected to have a low crowd.";
}

const result = document.getElementById("result");
result.className = "result " + cls;
document.getElementById("badge").textContent = level;
document.getElementById("desc").textContent = message;
document.getElementById("info").textContent =
  "Bus " + bus + " | " + time + " | " + day +
  " | Previous passengers: " + previous +
  " | Expected passengers: " + expected;

result.scrollIntoView({behavior:"smooth",block:"nearest"});
}
</script>
</body>
</html>
```